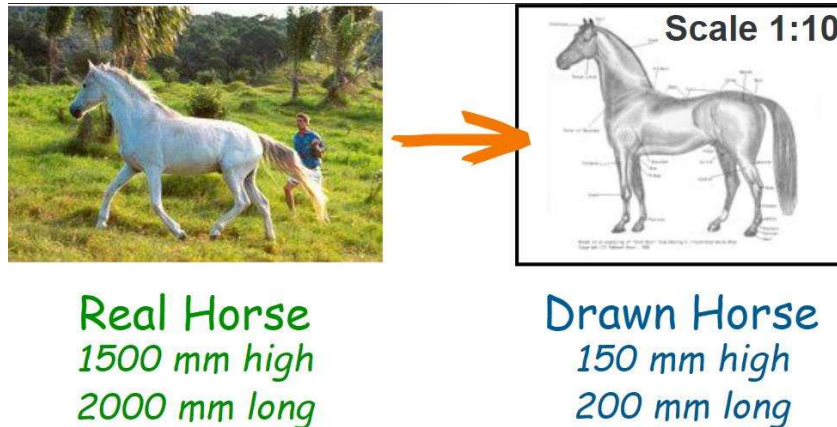


3.3 Scale

Scale is the ratio of the length in a drawing (or model) to the length of the real things.



In the drawing anything with the size of “1” would have a size of “10” in the real world, so a measurement of 150mm on the drawing would be 1500mm on real life.

Example 3.

A road 3km is represented by a line 8cm long. What is the length of a road 12cm long?

$$\begin{aligned} 8\text{cm} &= 3\text{km} \\ 8\text{cm} &= 3000\text{m} \\ \therefore 1\text{cm} &= ? \end{aligned}$$

Cross multiply $\frac{1 \times 3000}{8}$

$$\therefore 1\text{cm is } 375\text{m}$$

$$1\text{cm rep } 0.375\text{km}$$

$$\therefore 12\text{cm Will be } 0.375 \times 12$$

$$\underline{\underline{= 4.4\text{km}}}$$

Activity 5:

In groups measure the following and use the scale 1:100 to draw them on a paper.

- a. School Compound.
- b. Class room.
- c. Playing ground.

Exercise 4.

In groups, work out problems involving scale drawings.

1. A section of a tarmac road measures 8.5km, if it is drawn on a map it measures 17cm. What was the scale used?
2. A river measuring 5cm on a map has a length of 75km. What is the scale used on the map?
3. A map whose scale was 1:100000, an actual length of a right-angled plot of land measures 70m by 50m. What was the area of the triangular plot on the map?
4. On a map, 7.5cm represents 90km of the border of a certain county. What is the scale used?
5. A map is drawn to a scale of 1:15000. What is the distance in kilometres of a road which is 13.5cm on a map?
6. In scale drawing 1cm on a map represents an actual length of 25m. What area in the drawing will represent an actual area of 1 hectare?
7. A road is represented on a map by 3cm. What is the actual length of the road in kilometres if the scale used is 1:125000?